ECE 3441 Spring 2022
Exam 1

Version A

Please do not open booklet unless instructed to do so, or substantial penalty will result.

- Closed book, closed notes
- ONE 8.5 x 11” double-sided cheat sheet is allowed.
- Only four function calculators are allowed.
- Show all details of your work to obtain credit (The answer alone will not suffice).
- You may not talk during the quiz to others or to yourself.
- You have 80 minutes.

First Name Last Name, Peoplesoft ID, Date *DD/MM/YYYY*

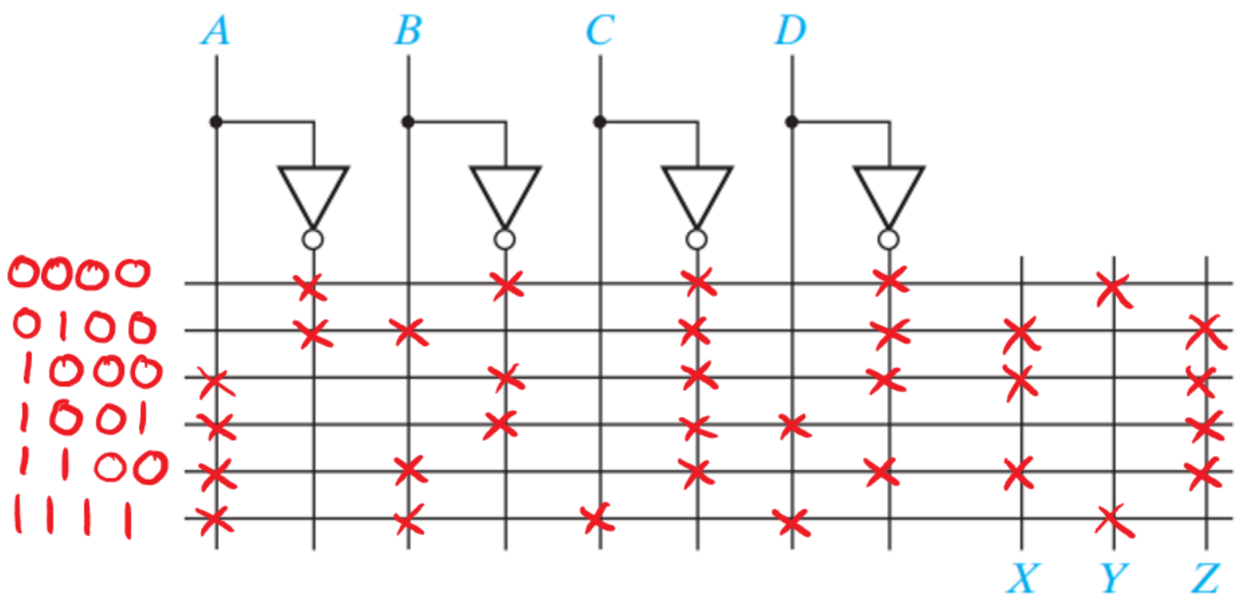
Question	Points	Maximum
1		20
2		15
3		15
4		20
5		20
6		20

Total: _______/100 possible points + 10 Bonus

Question 1:

Implement the following PLA to realize 3 number detectors (X, Y, Z). Each detector accepts a 4-bit binary input and works as follow:

- X outputs 1 if the decimal-equivalent value of the input is divisible by 4 (0 is not divisible by 4)
- Y outputs 1 if the input includes all 0s or all 1s.
- Z outputs 1 if the input contains the sequence 100, (*e.g. 100_ etc.*)



X $= \sum m(4,8,12)$

Y $= \sum m(0,15)$

Z $= \sum m(4,8,9,12)$

Question 1:

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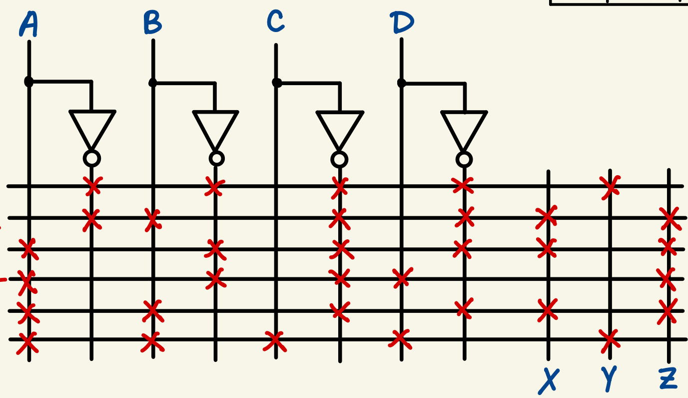
- Z outputs 1 if the input contains the sequence 100, (*e.g. 100_ etc.*)

AB \ CD	00	01	11	10
00	m_0 0000	m_1 0001	m_3 0011	m_2 0010
01	m_4 0100	m_5 0101	m_7 0111	m_6 0110
11	m_{12} 1100	m_{13} 1101	m_{15} 1111	m_{14} 1110
10	m_8 1000	m_9 1001	m_{11} 1011	m_{10} 1010

0	0000
4	0100
8	1000
9	1001
12	1100
15	1111



$X = \Sigma m(4, 8, 12)$

$Y = \Sigma m(0, 15)$

$Z = \Sigma m(4, 8, 9, 12)$

0, 4, 8, 9, 12, 15

Question 2:

Simplify the following expression using Boolean Algebra, identify each theorem you use.

$$G(A,B,C,D,E,F) = [(AB) \oplus (A+B)](B \oplus C) + (E \equiv F)(D \equiv E)(F \oplus D) + (A \oplus B)(B \equiv C)$$

You must use Boolean Algebra to receive any credit

$$[(AB) \oplus (A+B)](B \oplus C)$$
$$= [(AB)(A'B') + (A' + B')(A+B)](B \oplus C)$$
$$= (A'B + AB')(B \oplus C)$$
$$= (A \oplus B)(B \oplus C)$$

$$(E \equiv F)(D \equiv E)(F \oplus D)$$
$$= (E'F + EF')(D'E + DE')(D'F + DF')$$
$$= (DE'F + D'EF')(D'F + DF')$$
$$= 0$$

$$G = (A \oplus B)(B \oplus C) + 0 + (A \oplus B)(B \equiv C)$$
$$G = (A \oplus B)[(B \oplus C) + (B \equiv C)]$$
$$G = A \oplus B$$

Simplify the following expression using Boolean Algebra, identify each theorem you use.

$$G(A,B,C,D,E,F) = [(AB) \oplus (A+B)](B \oplus C) + (E \equiv F)(D \equiv E)(F \oplus D) + (A \oplus B)(B \equiv C)$$

You must use Boolean Algebra to receive any credit

Exclusive OR
$X \oplus Y = X'Y + XY'$

Equivalence
$(X \equiv Y) = XY + X'Y'$

Equivalence to Exclusive OR

$[(AB)'(A+B) + (AB)(A+B)'](B'C+BC') + (EF+E'F')\cancel{(DE+D'E')(F'D+FD')} + (A'B+AB')(BC+B'C')$

law of comp

$[(A'+B')(A+B) + (AB)(A'B')](B'C+BC') + (A'BC + A'\cancel{BB'}C' + A\cancel{BB'}C + AB'C')$

$[(\cancel{AA'} + A'B + AB' + \cancel{BB'}) + (\cancel{AA'BB'})](B'C+BC')$

$(A'B+AB')(B'C+BC') + A'BC + AB'C$

$A'\cancel{BB'}C + A'BC' + AB'C + A\cancel{BB'}C + A'BC + AB'C$

$B(A'C' + A'C) + B'(AC + AC)$

Question 3:

a) Convert the following number from binary to base 16: **11010110.1001_2**

b) Treating the binary numbers below as 2's complement. Perform the 6-bit addition and indicate overflow/no overflow.

110001 + 011011

$$\underbrace{1101}_{13 = D}\underbrace{0110}_{6}.\underbrace{1001}_{9}{}_2 = D6.9_{16}$$

$$\begin{array}{rrr} & {}^{1}1\,1\,0\,{}^{1}0\,{}^{1}0\,1 & -15 \\ + & 0\,1\,1\,0\,1\,1 & 27 \\ \hline (1) & 0\,0\,1\,1\,0\,0 & 12 \end{array}$$

no overflow

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b) Treating the binary numbers below as 2's complement. Perform the 6-bit addition and indicate overflow/no overflow.

110001 + 011011

$\underbrace{1101}_{13}\ \underbrace{0110}_{6}\ .\ \underbrace{1001}_{9}$

13=D

$\boxed{D6.9}$

5 4 3 2 1 0

16+8+2+1

24+2+1 = 27

$$\begin{array}{r} {}^{1}\ \ {}^{1}{}^{1}\ \ \\ 110001 \\ +\,011011 \\ \hline 1001100 \\ 6\,5\,4\,3\,2\,1\,0 \end{array}$$

→ 12

$$\begin{array}{r} 110001 \\ \hookrightarrow 001110 \\ +\qquad 1 \\ \hline 001111 \\ 5\,4\,3\,2\,1\,0 \end{array}$$

8 4 2 1

12 14 15

$$\begin{array}{r} -15 \\ +27 \\ \hline 12 \end{array}$$

There is no overflow

Question 4:

Given:

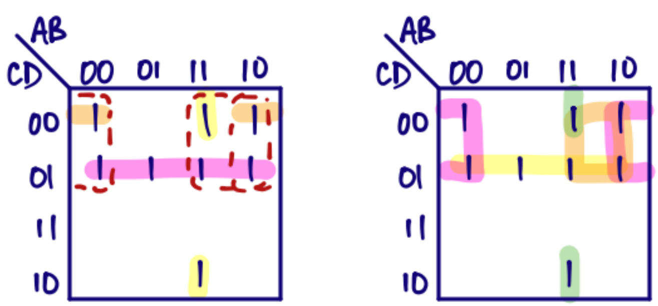
$$F_{1A}(A,B,C,D) = C'D + ABD' + B'C'D'$$

a) Use a K-Map to identify all 1 hazards.

b) Give the new function that has no 1 hazards.

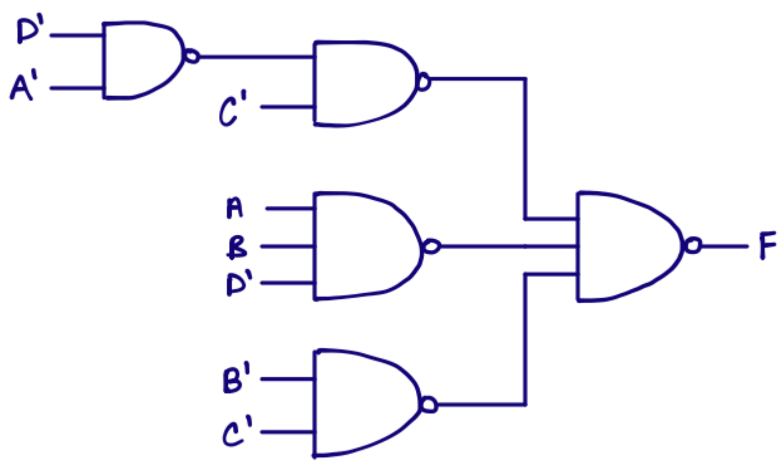
c) Realize the no hazard function using 3-input NAND gates.

$C'D + ABD' + B'C'D'$



no hazards: $C'D + AC' + ABD' + B'C'$

$C'(D + A) + ABD' + B'C'$



Given:

$$F_{1A}(A,B,C,D) = C'D + ABD' + B'C'D'$$

a) Use a K-Map to identify all 1 hazards.

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AB \ CD	00	01	11	10
00	m_0 1	m_1 1	m_3 0	m_2 0
01	m_4 0	m_5 1	m_7 0	m_6 0
11	m_{12} 1	m_{13} 1	m_{15} 0	m_{14} 1
10	m_8 1	m_9 1	m_{11} 0	m_{10} 0

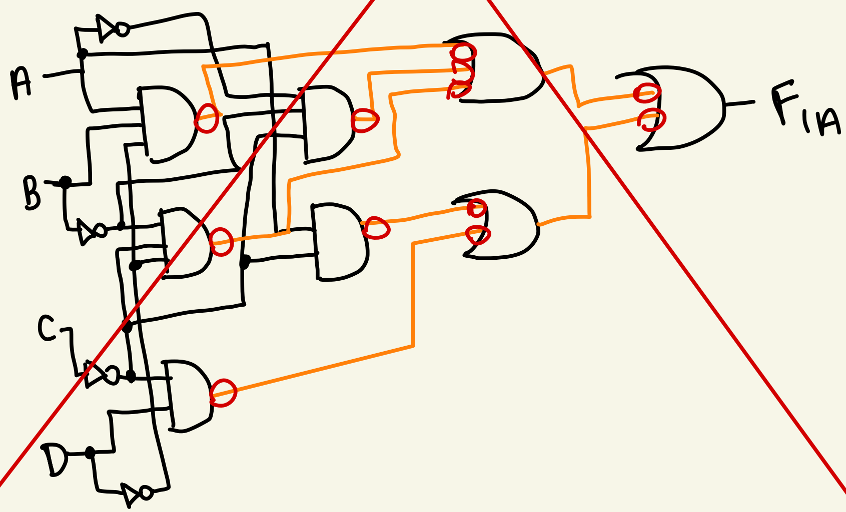
$C'D$ 01

ABD' 110

$B'C'D'$ 000

$F_{1A} = C'D + ABD' + B'C'D'$

$A'B'C' + AC'$



Question 5:

Given:

$$F_{2A}(W,X,Y,Z) = \prod M(1,5,11,12,14,15) \cdot \prod d(3,4,10)$$

a) Use K-maps to find the minimum product-of-sums (POS) of F_B.

b) Use K-maps to find the minimum sum-of-products (SOP) of F_B. (*You may use the same K-map as from part a*)

c) Realize the function using 3-input NOR gates.

YZ \ WX	00	01	11	10
00		X	0	
01	0	0		
11	X		0	0
10			0	X

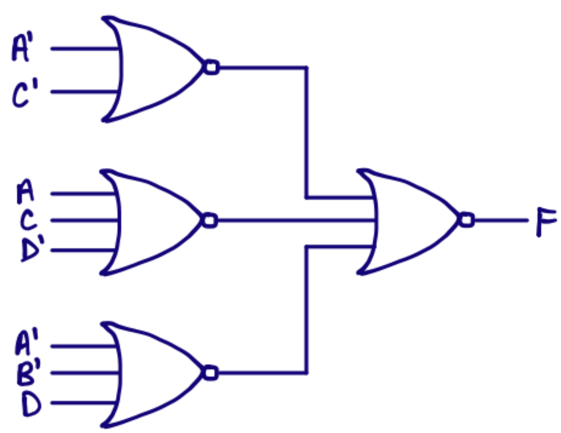
$F_B = (A' + C') \cdot (A + C + D') \cdot (B' + C + D)$

or

$F_B = (A' + C') \cdot (A + C + D') \cdot (A' + B' + D)$

YZ \ WX	00	01	11	10
00	1	X		1
01			1	1
11	X	1		
10	1	1		X

$F_B = B'D' + A'C + AC'D$



Given:

$$F_{2A}(W,X,Y,Z) = \prod M(1,5,11,12,14,15) \cdot \prod d(3,4,10)$$

a) Use K-maps to find the minimum product-of-sums (POS) of F_B.

b) Use K-maps to find the minimum sum-of-products (SOP) of F_B. (*You may use the same K-map as from part a*)

c) Realize the function using 3-input NOR gates.

POS

AB \ CD	00	01	11	10
00	m_0 0	m_1 1	m_3 X	m_2 0
01	m_4 X	m_5 1	m_7 0	m_6 0
11	m_{12} 1	m_{13} 0	m_{15} 1	m_{14} 1
10	m_8 0	m_9 0	m_{11} 1	m_{10} X

SOP

AB \ CD	00	01	11	10
00	m_0 0	m_1 1	m_3 X	m_2 0
01	m_4 X	m_5 1	m_7 0	m_6 0
11	m_{12} 1	m_{13} 0	m_{15} 1	m_{14} 1
10	m_8 0	m_9 0	m_{11} 1	m_{10} X

Question 6:

Given:

$$F_{3A}(A,B,C) = \sum m(0,4) + \sum d(1,5)$$

Derive a minimum SOP equation for F_A using Quine-McCluskey method.

Column I			Column II			Column III		
0	0	000	0	0, 1	00-	0	0,1,4,5	-0-
				0, 4	-00			
1	1	001	1	1, 5	-01	1		
	4	100		4, 5	10-			
2	5	101	2					
3								

Prime implicants:

0,1,4,5	-0-	B’

$$\Rightarrow F_A = B'$$

111
421

Given:

$$F_A(A, B, C) = \sum m(0,4) + \sum d(1,5)$$

Derive a minimum SOP equation for F_A using Quine-McCluskey method.

0	000	0,1 00- 0,4 -00	0,1,4,5 -0-
1	001	1,5 -01 1,4 10-	
4	100		
5	101		

$F_A = B'$